Open-SPEN in Pulseq

„Spatiotemporal Encoding“

Vendor-Agnostic Pulse Sequence
Programming with Pulseq II

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Motivation

Technical Limitations of EPI

- Geometric Distortions
  - Highly sensitive to B0 inhomogeneities
- Chemical-shift and susceptibility artefacts
- Eddy Currents warp the EPI "echo train"

Solution: SPatiotemporal ENcoding (SPEN)

- Direct spatial encoding in former low-bandwidth phase encoding direction
  - + Increased robustness towards off-resonance
  - - Inherently lower SNR
  - - high SAR (use of frequency swept RF pulses)
  - - increased minimal TE and prolonged Sequence duration

SPEN Principle: Quadratic Phase Profile

- SPEN correlates acquisition time directly to spatial position
- Key mechanism: Generation of a quadratic phase profile across Field of View (FOV)
  - Simultaneous application of:
    - A linearly frequency-swept (chirped) RF Pulse with amplitude-modulation
    - Constant magnetic field gradient
  - Gradient:
    - Spins at position y resonante with: $f_{res}(y) = f_{iso} + \gamma G y$
  - Chirped-RF:
    - Spins get excited / resonate at time-point t with: $f_{res}(t) = f_0 + Rt \rightarrow f_{res}(y) = f_0 + Rt_y$
    - Induced RF-phase: $\varphi_{RF}(t) = 2\pi(f_0 t + \frac{1}{2} R t^2) \rightarrow \varphi_{enc}(y) = 2\pi(f_0 t_y + \frac{1}{2} R {t_y}^2)$

$$R = \frac{BW}{T_{RF}}$$

SPEN Principle: Quadratic Phase Profile

- If $BW_{Gexc} = BW_{RF}$ s.t. $f_{Gexc}(y) = f_{RF}(y)$, then t_y is the time point where a spin at y (t_y) is excited

$$BW_{Gexc} = \gamma G \cdot \mathrm{FOV} \quad \text{and} \quad BW_{RF} = \frac{\mathrm{TBW}_{RF}}{T_{exc}}$$

$$f_{Gexc}(y) = f_{iso} + \gamma G_{exc} y \quad \text{and} \quad f_{RF}(y) = f_0 + R t_y$$

$$BW_{Gexc} = BW_{RF}$$

$$\gamma G_{exc} T_{exc} = \frac{\mathrm{TBW_{RF}}}{\mathrm{FOV}}$$

$$f_{iso} + \gamma G y = f_0 + R t_y$$

$$\rightarrow t_y = \frac{(f_{iso} - f_0) + \gamma G y}{R} = \left(\frac{\gamma G}{R}\right) y$$

$$\varphi_{enc}(y) = 2\pi\left(f_0 t_y + \frac{1}{2} R {t_y}^2\right)$$

$$= \underbrace{2\pi\gamma G \frac{f_0}{R} y}_{\text{linear}} + \underbrace{\pi \frac{\gamma^2 G^2}{R} y^2}_{\beta y^2 \text{ (quadratic)}}$$

$$\boldsymbol{\varphi_{enc}(y) = \beta y^2}$$

SPEN Principle: Conditions

Steepness

$$\gamma G_{exc} T_{exc} = \frac{\mathrm{TBW_{RF}}}{\mathrm{FOV}} = \frac{T_{exc} BW_{RF}}{\mathrm{FOV}}$$

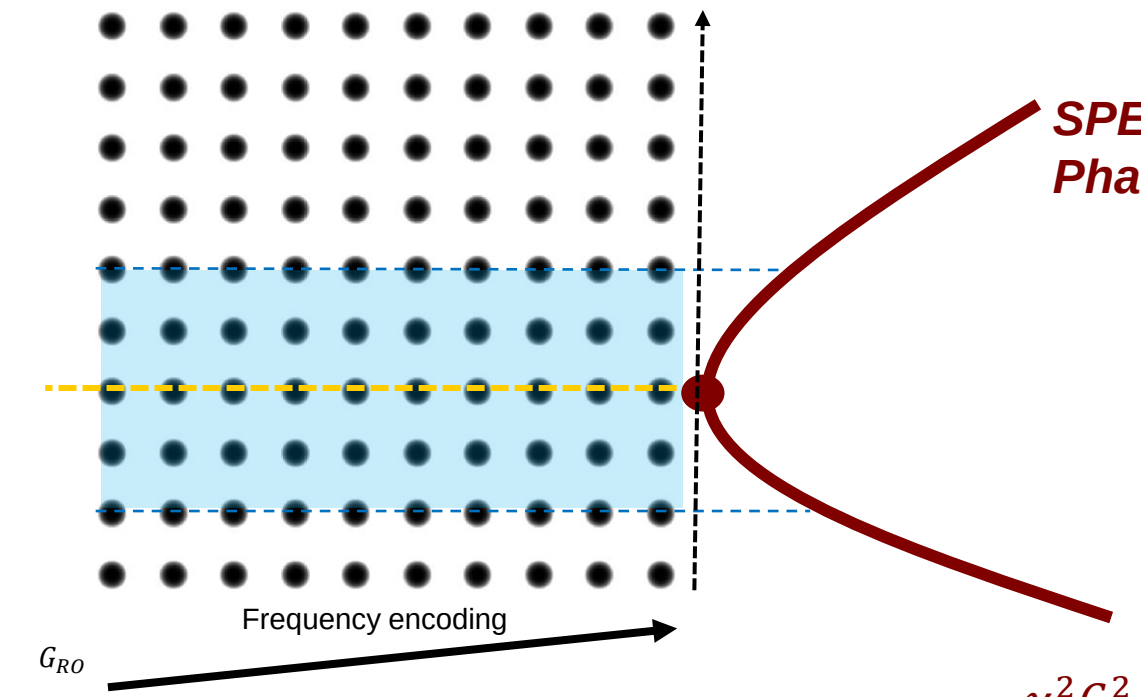
$$\gamma G_{exc} = \frac{BW_{RF}}{\mathrm{FOV}}$$

Acquisition

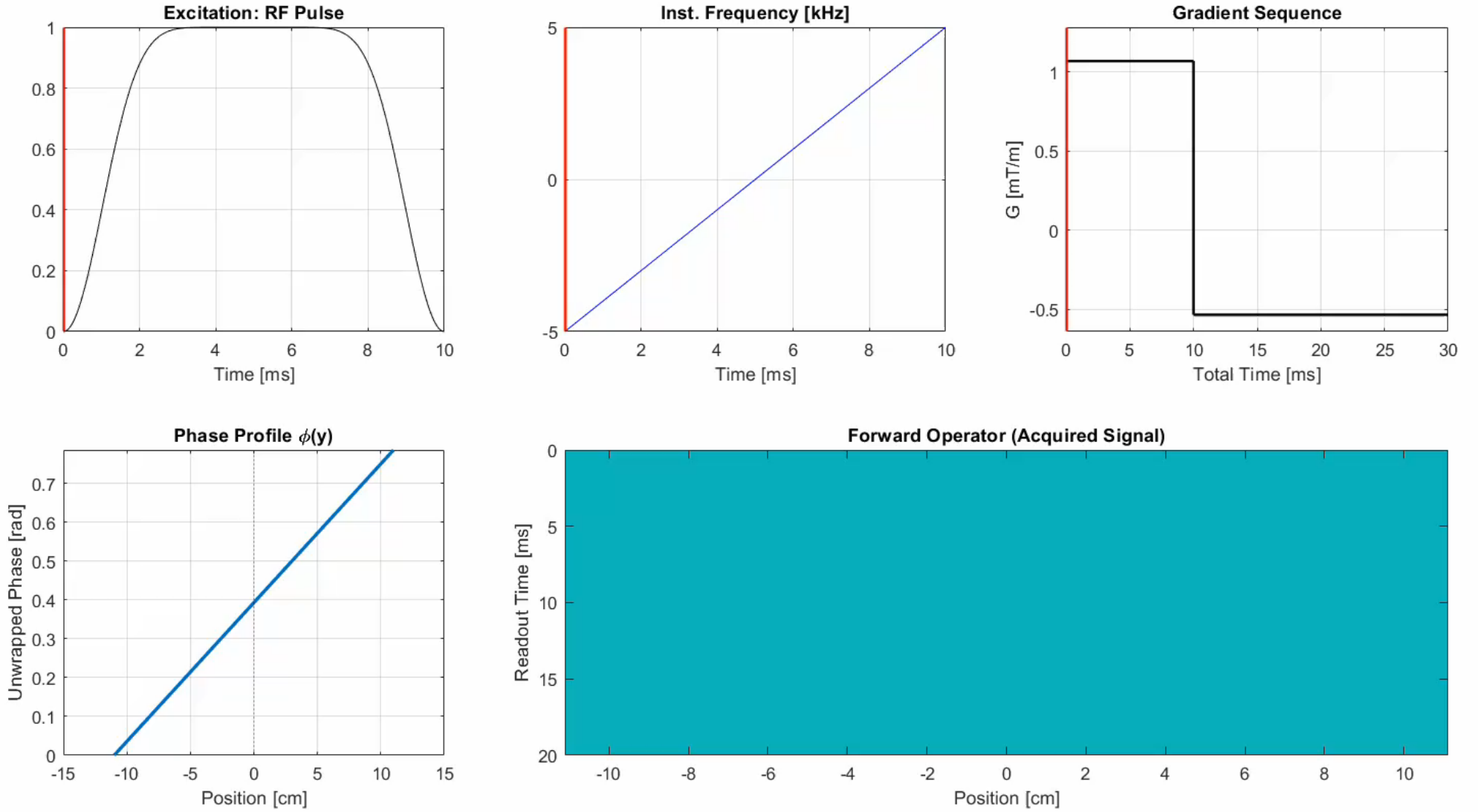
$$\gamma G_{exc} T_{exc} = \int_0^{T_{acq}} \gamma G_{acq} \, \mathrm{d}t$$

Stationary phase approximation

$$y_{stat} = -\frac{k_y(t)}{2\beta}$$

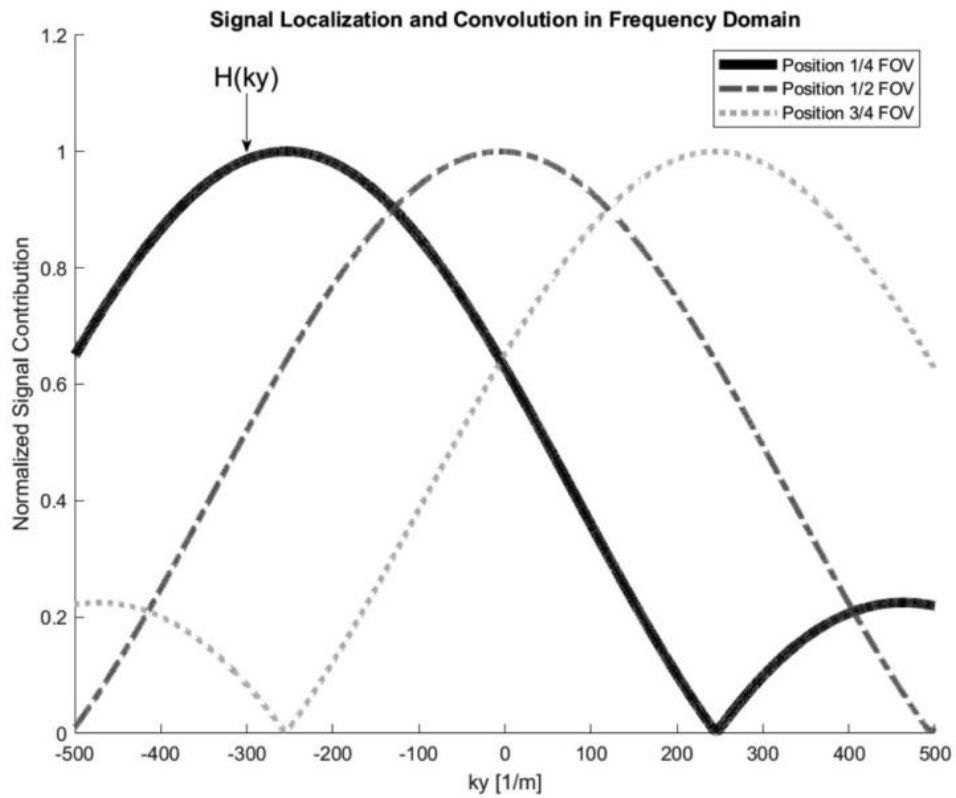


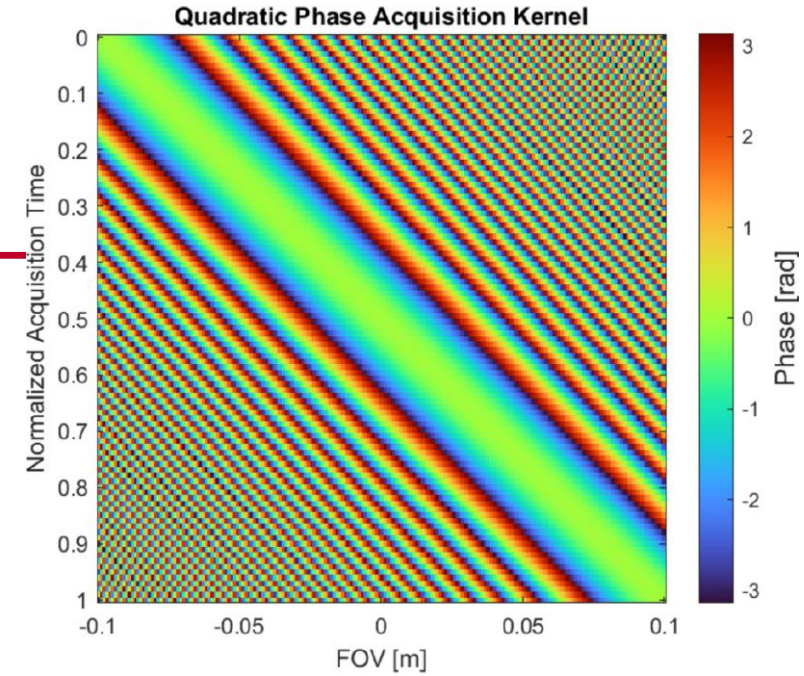
$$\varphi_{enc}(y) = \pi \frac{\gamma^2 G^2}{R} y^2$$

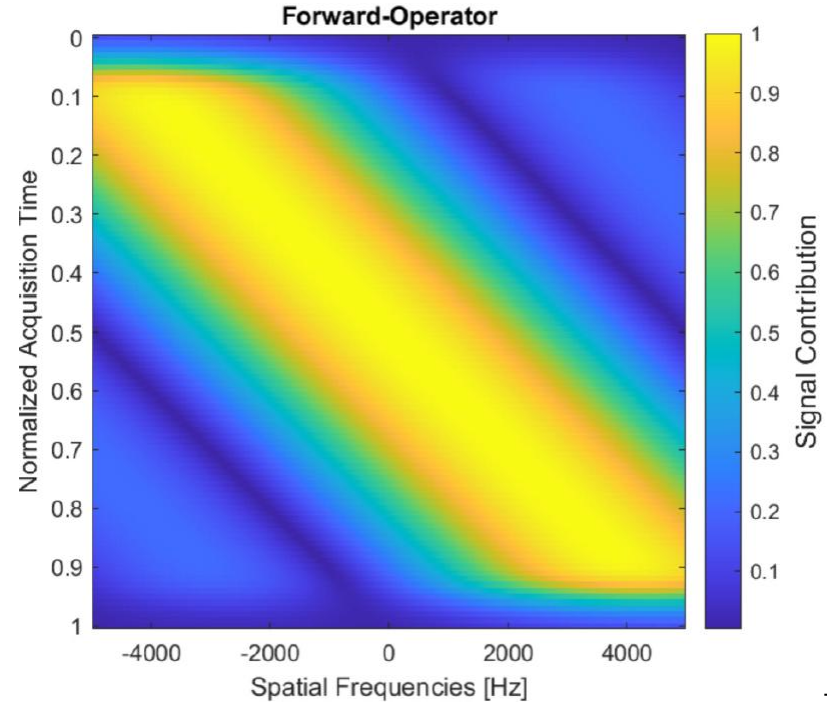

Excitation: RF Pulse
Time [ms]
Inst. Frequency [kHz]
Time [ms]
Gradient Sequence
G [mT/m]
Total Time [ms]
Phase Profile ϕ(y)
Unwrapped Phase [rad]
Position [cm]
Forward Operator (Acquired Signal)
Readout Time [ms]
Position [cm]

SPEN Principle: Forward Operator

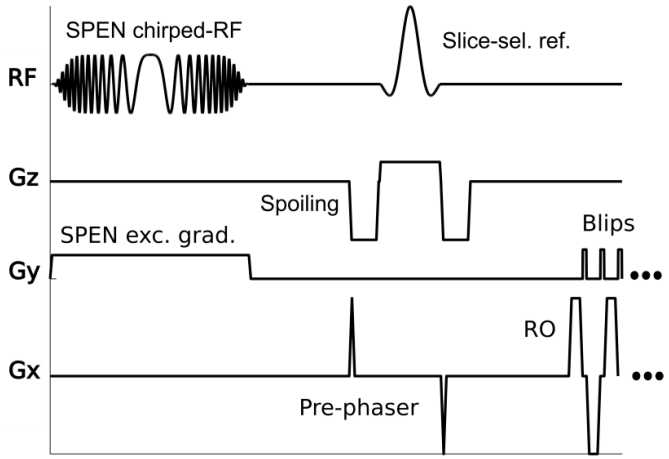
$$s(k_y) = \int_{-FOV/2}^{FOV/2} \rho(y) \cdot e^{ik_y y} \cdot e^{i\beta y^2} \, \mathrm{d}y = P(k_y) * H(k_y)$$

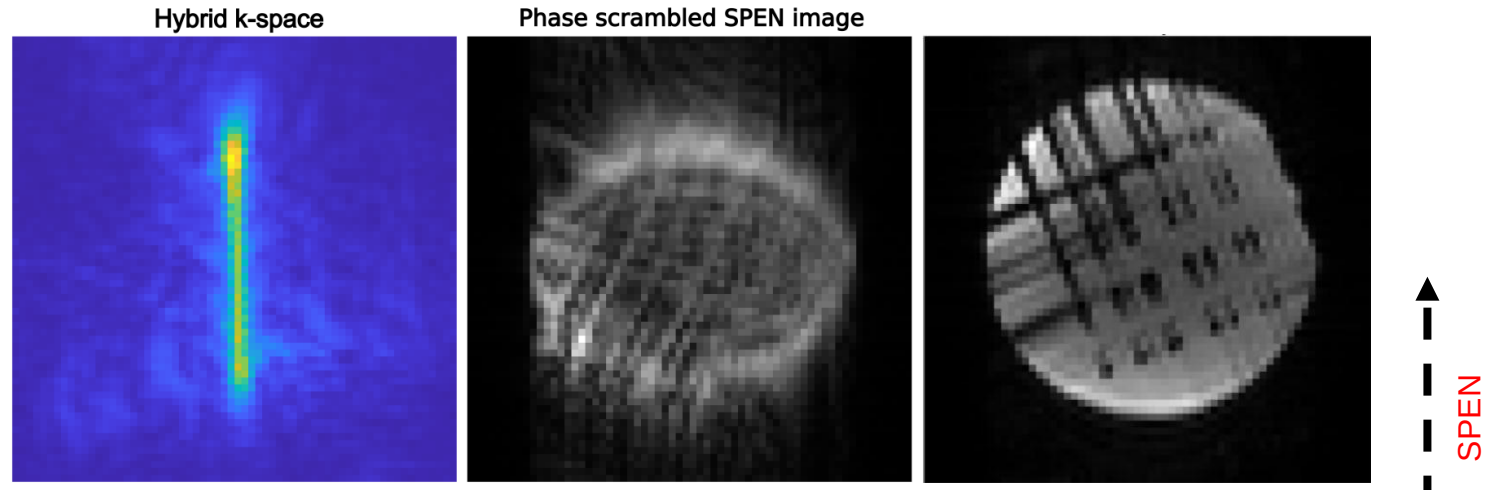






SPEN Principle: Forward Operator





$\mathcal{F}^{-1}$

SPEN-recon:
$H(k_y)^{-1}$

Freq. enc.

SPEN

Raw Data!!

Open-SPEN in Pulseq

- Example SPEN-SE-EPI and Multi-Slice DW-SPEN-SE-EPI Pulseq sequences
  - https://github.com/andih98/Open-SPEN.git
  - https://github.com/andih98

- More in-depth explanation of SPEN-Theory: https://doi.org/10.1002/mrm.70277

RESEARCH ARTICLE OPEN ACCESS

Spatiotemporal Encoding With Nonlinear Gradient Hardware Using Pulseq: From Principles to Practical Demonstration

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Open-SPEN in Pulseq: Sequence assembly

SPEN-SE-EPI

SPEN excitation

- Chirped RF pulse
- SPEN gradient G_{exc}

$$\gamma G_{exc} = \frac{BW_{RF}}{\mathrm{FOV}}$$

SE-generation

- Slice selective 180°-refocusing
- Pre-phasing gradients

Modified EPI Readout

- Bipolar RO-gradients
- Phase-Blips fulfilling:

$$\gamma G_{exc} T_{exc} = \int_0^{T_{acq}} \gamma G_{acq} \, \mathrm{d}t$$

Open-SPEN in Pulseq

Interactive Pulseq Sequence Assembly

„SPEN-SE-EPI“

(chirped-RF exc.)